11. Write a Java Program to Input an Integer and Check Whether It Is Even or Odd Using if-else

```
import java.util.Scanner;


public class EvenOdd {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    System.out.print("Enter an integer: ");
    int num = sc.nextInt();


    if (num % 2 == 0) {
      System.out.println("Even Number");
    } else {
      System.out.println("Odd Number");
    }
    sc.close();
  }
}
```

12. Write a Java program to input a number and check whether it is positive, negative, or zero.

```
import java.util.Scanner;

public class PositiveNegativeZero {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");
```

```java
        int num = sc.nextInt();

        if (num > 0) {
            System.out.println("Positive Number");
        } else if (num < 0) {
            System.out.println("Negative Number");
        } else {
            System.out.println("Zero");
        }

        sc.close();
    }
}
```

13. Write a Java program to input 3 integers and find the largest using if-else ladder.

```java
import java.util.Scanner;

public class LargestNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
```

```java
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number = " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number = " + b);
        } else {
            System.out.println("Largest number = " + c);
        }

        sc.close();
    }
}
```

14. Write a Java program to input a year and check whether it is a leap year or not.

```java
import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a year: ");
        int year = sc.nextInt();

        if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
            System.out.println(year + " is a Leap Year");
        } else {
            System.out.println(year + " is not a Leap Year");
        }
```

```
      sc.close();
   }
}
```

15. Write a Java program to input a character and check whether it is a vowel or consonant using switch.

```
import java.util.Scanner;

public class VowelConsonant {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a character: ");
      char ch = sc.next().charAt(0);

      switch(ch) {
         case 'a':
         case 'e':
         case 'i':
         case 'o':
         case 'u':
         case 'A':
         case 'E':
         case 'I':
         case 'O':
         case 'U':
            System.out.println("Vowel");
            break;

         default:
            System.out.println("Consonant");
      }

      sc.close();
   }
}
```

16. Write a Java program to create a menu-driven calculator using switch for +- * / %.

```
import java.util.Scanner;

class Calculator {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int a, b;
    char op;

    System.out.print("Enter first number: ");
    a = sc.nextInt();

    System.out.print("Enter second number: ");
    b = sc.nextInt();

    System.out.print("Enter operator (+, -, *, /, %): ");
    op = sc.next().charAt(0);

    switch(op) {
      case '+':
        System.out.println("Result = " + (a + b));
        break;
```

```java
        case '-':
          System.out.println("Result = " + (a - b));
          break;

        case '*':
          System.out.println("Result = " + (a * b));
          break;

        case '/':
          System.out.println("Result = " + (a / b));
          break;

        case '%':
          System.out.println("Result = " + (a % b));
          break;

        default:
          System.out.println("Invalid Operator");
      }

      System.out.println("\nAdarsha Raj Luitel");

      sc.close();
    }
}
```

17. Write a Java program to print numbers from 1 to N using a for loop

```java
import java.util.Scanner;

class PrintNumbers {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n;

    System.out.print("Enter N: ");
    n = sc.nextInt();

    for(int i = 1; i <= n; i++) {
      System.out.println(i);
    }

    System.out.println("\nAdarsha Raj Luitel");

    sc.close();
  }
}
```

18. Write a Java program to print numbers from N to 1 using a while loop.

```java
import java.util.Scanner;

class ReverseNumbers {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n;

    System.out.print("Enter N: ");
    n = sc.nextInt();

    while(n >= 1) {
      System.out.println(n);
```

```
      n--;
    }

    System.out.println("\nAdarsha Raj Luitel");

    sc.close();
  }
}
```

19. Write a Java program to find the sum of first N natural numbers using a loop

```
import java.util.Scanner;

class SumNatural {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n, sum = 0;

    System.out.print("Enter N: ");
    n = sc.nextInt();

    for(int i = 1; i <= n; i++) {
      sum = sum + i;
    }

    System.out.println("Sum = " + sum);

    System.out.println("\nAdarsha Raj Luitel");

    sc.close();
  }
}
```

20. Write a Java program to find the factorial of a number using a for loop.

```
import java.util.Scanner;
```

```
class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n;
        long fact = 1;

        System.out.print("Enter a number: ");
        n = sc.nextInt();

        for(int i = 1; i <= n; i++) {
            fact = fact * i;
        }

        System.out.println("Factorial = " + fact);

        System.out.println("\nAdarsha Raj Luitel");

        sc.close();
    }
}
```

21. Write a Java program to check whether a number is prime or not.

```
import java.util.Scanner;

class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, i;
        boolean prime = true;

        System.out.print("Enter a number: ");
        n = sc.nextInt();
```

```java
        if(n <= 1) {
            prime = false;
        } else {
            for(i = 2; i < n; i++) {
                if(n % i == 0) {
                    prime = false;
                    break;
                }
            }
        }

        if(prime)
            System.out.println("Prime Number");
        else
            System.out.println("Not a Prime Number");

        System.out.println("\nAdarsha Raj Luitel");

        sc.close();
    }
}
```

22. Write a Java program to reverse a number (Example: 123 → 321) using a loop.

```java
import java.util.Scanner;

class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, rev = 0, rem;

        System.out.print("Enter a number: ");
        n = sc.nextInt();

        while(n != 0) {
            rem = n % 10;
```

```java
            rev = rev * 10 + rem;
            n = n / 10;
        }

        System.out.println("Reversed Number = " + rev);

        System.out.println("\nAdarsha Raj Luitel");

        sc.close();
    }
}
```

23. Write a Java program to check whether a number is a palindrome or not.

```java
import java.util.Scanner;

class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, temp, rem, rev = 0;

        System.out.print("Enter a number: ");
        n = sc.nextInt();

        temp = n;

        while(n != 0) {
            rem = n % 10;
            rev = rev * 10 + rem;
            n = n / 10;
        }

        if(temp == rev)
            System.out.println("Palindrome Number");
        else
            System.out.println("Not a Palindrome Number");
```

```
        System.out.println("\nAdarsha Raj Luitel");

        sc.close();
    }
}
```

24. Write a Java program to input an integer and count the number of digits in it.

```
import java.util.Scanner;

class CountDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, count = 0;

        System.out.print("Enter an integer: ");
        n = sc.nextInt();

        while(n != 0) {
            count++;
            n = n / 10;
        }

        System.out.println("Number of digits = " + count);

        System.out.println("\nAdarsha Raj Luitel");

        sc.close();
    }
}
```

25. Write a Java program to input a number and check whether it is an Armstrong number or not.

```
import java.util.Scanner;

class Armstrong { public static void main(String[] args) { Scanner sc = new
Scanner(System.in);

  int n, temp, rem, sum = 0;

   System.out.print("Enter a number: ");
   n = sc.nextInt();

   temp = n;

   while(n != 0) {
      rem = n % 10;
      sum = sum + (rem * rem * rem);
      n = n / 10;
   }

   if(temp == sum)
      System.out.println("Armstrong Number");
   else
      System.out.println("Not an Armstrong Number");

   System.out.println("\nAdarsha Raj Luitel");

   sc.close();
}


}
```